Clustering, $m = 5$ labelled windows per cluster

Table 1: Clustering, L1 (per-house activities, resident id removed), rare classes masked; $m = 5$. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
$K = 20$ clusters (100 labels)									
k-means	Sensor hist.	**0.74**	**0.50**	**0.87**	**0.63**	0.28	0.29	0.63	**0.47**
k-means	Sensor+ToD hist.	0.71	0.35	0.79	0.37	**0.78**	**0.52**	**0.76**	0.41
$K = 50$ clusters (250 labels)									
k-means	Sensor hist.	**0.75**	**0.55**	**0.87**	**0.61**	0.30	0.28	0.64	0.48
k-means	Sensor+ToD hist.	0.72	0.42	0.86	0.57	**0.82**	**0.62**	**0.80**	**0.54**

Table 2: Clustering, L1, all classes; $m = 5$. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
$K = 20$ clusters (100 labels)									
k-means	Sensor hist.	**0.73**	**0.37**	**0.87**	**0.44**	0.28	0.26	0.63	**0.36**
k-means	Sensor+ToD hist.	0.71	0.26	0.79	0.26	**0.78**	**0.47**	**0.76**	0.33
$K = 50$ clusters (250 labels)									
k-means	Sensor hist.	**0.74**	**0.40**	**0.87**	**0.43**	0.30	0.26	0.64	0.36
k-means	Sensor+ToD hist.	0.71	0.31	0.85	0.41	**0.82**	**0.56**	**0.80**	**0.43**

Table 3: Clustering, L2 (TDOST merged classes), all classes — comparable to TDOST / DeepCASAS; $m = 5$. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
$K = 20$ clusters (100 labels)									
k-means	Sensor hist.	**0.76**	**0.42**	**0.87**	**0.48**	0.74	0.30	**0.79**	**0.40**
k-means	Sensor+ToD hist.	0.72	0.30	0.79	0.31	**0.84**	**0.40**	0.78	0.34
$K = 50$ clusters (250 labels)									
k-means	Sensor hist.	**0.75**	**0.41**	**0.88**	**0.48**	0.75	0.30	0.79	0.40
k-means	Sensor+ToD hist.	0.74	0.37	0.86	**0.47**	**0.87**	**0.43**	**0.82**	**0.43**

Clustering, $m = 1$ labelled windows per cluster

Table 4: Clustering, L1 (per-house activities, resident id removed), rare classes masked; $m = 1$. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
$K = 20$ clusters (20 labels)									
k-means	Sensor hist.	**0.65**	**0.42**	**0.87**	**0.64**	0.27	0.30	0.60	**0.46**
k-means	Sensor+ToD hist.	0.61	0.35	0.73	0.35	**0.75**	**0.45**	**0.70**	0.39
$K = 50$ clusters (50 labels)									
k-means	Sensor hist.	**0.74**	**0.45**	**0.87**	**0.59**	0.28	0.27	0.63	0.44
k-means	Sensor+ToD hist.	0.66	0.40	0.81	0.56	**0.66**	**0.48**	**0.71**	**0.48**

Table 5: Clustering, L1, all classes; $m = 1$. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
$K = 20$ clusters (20 labels)									
k-means	Sensor hist.	**0.65**	**0.31**	**0.87**	**0.49**	0.27	0.27	0.60	**0.36**
k-means	Sensor+ToD hist.	0.60	0.26	0.73	0.24	**0.75**	**0.41**	**0.70**	0.30
$K = 50$ clusters (50 labels)									
k-means	Sensor hist.	**0.74**	**0.38**	**0.86**	**0.46**	0.28	0.24	0.63	0.36
k-means	Sensor+ToD hist.	0.66	0.30	0.81	0.40	**0.66**	**0.44**	**0.71**	**0.38**

Table 6: Clustering, L2 (TDOST merged classes), all classes — comparable to TDOST / DeepCASAS; $m = 1$. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
$K = 20$ clusters (20 labels)									
k-means	Sensor hist.	**0.66**	**0.34**	**0.88**	**0.54**	0.74	0.32	**0.76**	**0.40**
k-means	Sensor+ToD hist.	0.61	0.26	0.73	0.29	**0.82**	**0.41**	0.72	0.32
$K = 50$ clusters (50 labels)									
k-means	Sensor hist.	**0.74**	**0.37**	**0.87**	**0.53**	0.68	0.27	**0.76**	**0.39**
k-means	Sensor+ToD hist.	0.66	0.30	0.81	0.41	**0.74**	**0.40**	0.74	0.37

Clustering, $m = all$ labelled windows per cluster

Table 7: Clustering, L1 (per-house activities, resident id removed), rare classes masked; $m = all$. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
$K = 20$ *clusters*									
k-means	Sensor hist.	**0.79**	**0.58**	**0.86**	**0.59**	0.32	0.31	0.66	**0.49**
k-means	Sensor+ToD hist.	0.72	0.39	0.79	0.37	**0.77**	**0.49**	**0.76**	0.41
$K = 50$ *clusters*									
k-means	Sensor hist.	**0.80**	**0.60**	**0.87**	**0.61**	0.35	0.33	0.68	0.51
k-means	Sensor+ToD hist.	0.78	0.53	0.86	0.60	**0.81**	**0.61**	**0.82**	**0.58**

Table 8: Clustering, L1, all classes; $m = all$. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
$K = 20$ *clusters*									
k-means	Sensor hist.	**0.79**	**0.42**	**0.86**	**0.41**	0.32	0.28	0.66	**0.37**
k-means	Sensor+ToD hist.	0.71	0.29	0.79	0.26	**0.77**	**0.44**	**0.76**	0.33
$K = 50$ *clusters*									
k-means	Sensor hist.	**0.80**	**0.49**	**0.87**	**0.43**	0.35	0.30	0.67	0.41
k-means	Sensor+ToD hist.	0.78	0.39	0.86	0.42	**0.80**	**0.56**	**0.82**	**0.46**

Table 9: Clustering, L2 (TDOST merged classes), all classes — comparable to TDOST / DeepCASAS; $m = all$. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
$K = 20$ *clusters*									
k-means	Sensor hist.	**0.79**	**0.45**	**0.87**	**0.48**	0.74	0.30	**0.80**	**0.41**
k-means	Sensor+ToD hist.	0.73	0.31	0.79	0.31	**0.84**	**0.40**	0.79	0.34
$K = 50$ *clusters*									
k-means	Sensor hist.	**0.81**	**0.46**	**0.88**	**0.49**	0.79	0.36	0.82	0.44
k-means	Sensor+ToD hist.	0.79	0.42	0.87	0.48	**0.87**	**0.43**	**0.84**	**0.45**

Linear probes (logistic regression on frozen representations)

Table 10: Linear probes, L1 (per-house activities, resident id removed), rare classes masked. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
100 random labelled windows									
Linear probe	Sensor hist.	**0.71**	0.38	**0.82**	**0.42**	0.29	0.19	0.61	0.33
Linear probe	Sensor+ToD hist.	**0.71**	**0.40**	**0.81**	**0.42**	**0.60**	**0.34**	**0.71**	**0.39**
250 random labelled windows									
Linear probe	Sensor hist.	**0.74**	0.48	**0.85**	**0.55**	0.32	0.27	0.64	0.43
Linear probe	Sensor+ToD hist.	**0.75**	**0.49**	**0.84**	0.53	**0.68**	**0.49**	**0.75**	**0.50**
all labelled windows (max 20k)									
Linear probe	Sensor hist.	0.81	0.62	**0.89**	0.63	0.38	0.38	0.69	0.54
Linear probe	Sensor+ToD hist.	**0.83**	**0.65**	**0.89**	**0.64**	**0.83**	**0.65**	**0.85**	**0.65**

Table 11: Linear probes, L1, all classes. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
100 random labelled windows									
Linear probe	Sensor hist.	**0.71**	0.30	**0.82**	**0.29**	0.29	0.18	0.60	0.26
Linear probe	Sensor+ToD hist.	**0.71**	**0.31**	**0.81**	**0.29**	**0.60**	**0.32**	**0.71**	**0.31**
250 random labelled windows									
Linear probe	Sensor hist.	**0.74**	0.37	**0.85**	**0.41**	0.32	0.25	0.63	0.34
Linear probe	Sensor+ToD hist.	**0.74**	**0.38**	**0.84**	0.40	**0.67**	**0.45**	**0.75**	**0.41**
all labelled windows (max 20k)									
Linear probe	Sensor hist.	0.81	**0.52**	**0.89**	0.48	0.38	0.38	0.69	0.46
Linear probe	Sensor+ToD hist.	**0.83**	0.50	**0.89**	**0.53**	**0.83**	**0.64**	**0.85**	**0.56**

Table 12: Linear probes, L2 (TDOST merged classes), all classes — comparable to TDOST / DeepCASAS. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	Milan		Aruba		Cairo		Average	
		W	M	W	M	W	M	W	M
100 random labelled windows									
Linear probe	Sensor hist.	**0.72**	0.30	**0.82**	**0.36**	0.70	0.27	0.75	0.31
Linear probe	Sensor+ToD hist.	**0.72**	**0.32**	**0.82**	**0.36**	**0.75**	**0.32**	**0.76**	**0.33**
250 random labelled windows									
Linear probe	Sensor hist.	**0.75**	0.37	**0.85**	**0.47**	0.73	0.31	0.78	0.38
Linear probe	Sensor+ToD hist.	**0.75**	**0.38**	**0.85**	0.46	**0.80**	**0.40**	**0.80**	**0.41**
all labelled windows (max 20k)									
Linear probe	Sensor hist.	0.81	0.49	**0.89**	0.55	0.80	0.43	0.84	0.49
Linear probe	Sensor+ToD hist.	**0.83**	**0.54**	**0.90**	**0.58**	**0.90**	**0.55**	**0.88**	**0.56**

Linear probe vs clustering at matched label budgets

Table 13: Probe vs clustering at matched label budgets, L1 (per-house activities, resident id removed), rare classes masked. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	K20, m5		100 labels		K50, m5		250 labels		K50, m all		all labels	
		W	M	W	M	W	M	W	M	W	M	W	M
three-house average; clustering columns = k-means / SCAN / DEC, probe columns = same representation													
k-means / probe	Sensor hist.	0.63	**0.47**	0.61	0.33	0.64	0.48	0.64	0.43	0.68	0.51	0.69	0.54
k-means / probe	Sensor+ToD hist.	**0.76**	0.41	**0.71**	**0.39**	**0.80**	**0.54**	**0.75**	**0.50**	**0.82**	**0.58**	**0.85**	**0.65**
k-means / probe	MLM	–	–	–	–	–	–	–	–	–	–	–	–
k-means / probe	MLM-time	–	–	–	–	–	–	–	–	–	–	–	–
k-means / probe	MLM + ToD	–	–	–	–	–	–	–	–	–	–	–	–
k-means / probe	MLM-time + ToD	–	–	–	–	–	–	–	–	–	–	–	–
SCAN	MLM-time	–	–	–	–	–	–	–	–	–	–	–	–
SCAN	MLM-time + ToD	–	–	–	–	–	–	–	–	–	–	–	–
DEC	MLM-time	–	–	–	–	–	–	–	–	–	–	–	–
DEC	MLM-time + ToD	–	–	–	–	–	–	–	–	–	–	–	–

Table 14: Probe vs clustering at matched label budgets, L1, all classes. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	K20, m5		100 labels		K50, m5		250 labels		K50, m all		all labels	
		W	M	W	M	W	M	W	M	W	M	W	M
three-house average; clustering columns = k-means / SCAN / DEC, probe columns = same representation													
k-means / probe	Sensor hist.	0.63	**0.36**	0.60	0.26	0.64	0.36	0.63	0.34	0.67	0.41	0.69	0.46
k-means / probe	Sensor+ToD hist.	**0.76**	0.33	**0.71**	**0.31**	**0.80**	**0.43**	**0.75**	**0.41**	**0.82**	**0.46**	**0.85**	**0.56**
k-means / probe	MLM	–	–	–	–	–	–	–	–	–	–	–	–
k-means / probe	MLM-time	–	–	–	–	–	–	–	–	–	–	–	–
k-means / probe	MLM + ToD	–	–	–	–	–	–	–	–	–	–	–	–
k-means / probe	MLM-time + ToD	–	–	–	–	–	–	–	–	–	–	–	–
SCAN	MLM-time	–	–	–	–	–	–	–	–	–	–	–	–
SCAN	MLM-time + ToD	–	–	–	–	–	–	–	–	–	–	–	–
DEC	MLM-time	–	–	–	–	–	–	–	–	–	–	–	–
DEC	MLM-time + ToD	–	–	–	–	–	–	–	–	–	–	–	–

Table 15: Probe vs clustering at matched label budgets, L2 (TDOST merged classes), all classes — comparable to TDOST / DeepCASAS. W/M = weighted/macro F1; bold = best in column within a block, shaded = within 0.03 of it.

Method	Repr.	K20, m5		100 labels		K50, m5		250 labels		K50, m all		all labels	
		W	M	W	M	W	M	W	M	W	M	W	M
three-house average; clustering columns = k-means / SCAN / DEC, probe columns = same representation													
k-means / probe	Sensor hist.	**0.79**	**0.40**	0.75	0.31	0.79	0.40	0.78	0.38	0.82	0.44	0.84	0.49
k-means / probe	Sensor+ToD hist.	0.78	0.34	**0.76**	**0.33**	**0.82**	**0.43**	**0.80**	**0.41**	**0.84**	**0.45**	**0.88**	**0.56**
k-means / probe	MLM	–	–	–	–	–	–	–	–	–	–	–	–
k-means / probe	MLM-time	–	–	–	–	–	–	–	–	–	–	–	–
k-means / probe	MLM + ToD	–	–	–	–	–	–	–	–	–	–	–	–
k-means / probe	MLM-time + ToD	–	–	–	–	–	–	–	–	–	–	–	–
SCAN	MLM-time	–	–	–	–	–	–	–	–	–	–	–	–
SCAN	MLM-time + ToD	–	–	–	–	–	–	–	–	–	–	–	–
DEC	MLM-time	–	–	–	–	–	–	–	–	–	–	–	–
DEC	MLM-time + ToD	–	–	–	–	–	–	–	–	–	–	–	–